

Medical Insurance Claim Fraud Detection with Leakage-Aware Traditional Machine Learning

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Abstract—This paper presents a reproducible classical machine-learning pipeline for medical insurance claim fraud screening. Using a repository-provided workbook of 4,500 claims with a 6.0% fraud rate, we construct financial and temporal screening features, exclude identifier-like and potentially post-decision fields, and fit all preprocessing on a stratified training partition. Twenty model configurations spanning linear, probabilistic, tree, boosting, instance, neural-baseline, and ensemble families are compared using fraud-aware metrics. Validation F2 selects the operating threshold and ranks models, while PR-AUC provides a complementary ranking view under class imbalance. The selected model is refit once on training plus validation rows before a locked test evaluation. The artifact chain includes curves, calibration, permutation importance, fairness slices, serialized model state, a presentation, and two PDF reports. We present the result as a decision-support baseline rather than an autonomous denial system. The supplied snapshot lacks policy, hospital-tier, document, and verified Indian geography fields; consequently, the findings are limited to the stated dataset and protocol and motivate later deep-learning and agentic extensions.

Keywords—fraud detection, health insurance, imbalanced classification, explainable AI, India

1. INTRODUCTION

Healthcare insurance fraud can appear as inflated charges, duplicate submissions, services not rendered, or inconsistent claim records. Manual investigation is necessary but expensive at scale. A machine-learning screening layer can prioritize records for review, provided that its errors, threshold, and limitations are visible.

This work addresses a constrained binary task: given a supplied structured claim record, estimate whether its label is Fraud or Legitimate. The contribution is an end-to-end, auditable baseline rather than a claim of population-level superiority. The implementation separates data validation, feature engineering, preprocessing, model fitting, evaluation, explainability, and document generation.

2. RELATED WORK

Statistical fraud detection has long emphasized the trade-off between missed fraud and investigation burden [1]. Anomaly detection surveys motivate deviation-based signals [11], while SMOTE [2] and cost-sensitive learning address imbalance. Random forests [3], gradient boosting [4], margin methods [5], and scikit-learn's unified evaluation framework [15] provide complementary inductive biases for mixed tabular data. SHAP-style explanations [10] motivate the broader interpretability direction, but this baseline uses permutation importance to keep the evidence chain lightweight. Healthcare fraud studies emphasize label quality, coding complexity, provider concentration, and temporal shift [12]–[14].

3. DATA AND METHOD

The workbook contains 4,500 rows and 19 source columns. Fraud accounts for 6.0%. Claim amount, income, age, dates, low-cardinality categories, claim type, submission method, specialty, and a supplied cluster field are available. IDs and near-unique diagnosis, procedure, and location values are excluded. Claim status is excluded because it may be observed after adjudication. The target is encoded as 1 for fraud and 0 for legitimate.

We use a 70/15/15 stratified split with seed 42. Numeric fields receive median imputation and standard scaling; categoricals receive most-frequent

imputation and unknown-safe one-hot encoding. The transformer is fitted on training rows only. Feature engineering adds log amounts, claim-to-income ratios, calendar terms, cyclic month encodings, age bands, and screening-time interactions.

The model zoo includes majority, logistic L1/L2, decision tree, random forest, extra trees, gradient and histogram gradient boosting, AdaBoost, RBF-SVM, KNN, Gaussian and Bernoulli naive Bayes, LDA, QDA, MLP, calibrated ridge, passive-aggressive, voting, and stacking. Searches use stratified three-fold average precision with small declared budgets. Validation thresholds maximize F2 subject to a preferred precision floor.

4. EVALUATION

We report accuracy, balanced accuracy, precision, recall, F1, F2, ROC-AUC, PR-AUC, MCC, Cohen's kappa, Brier score, log loss, specificity, confusion matrices, training time, prediction latency, and artifact size. PR-AUC is emphasized under imbalance; F2 captures the operational preference for recall. The selected model is Soft voting ensemble with validation F2 1.0000, validation PR-AUC 1.0000, and threshold 0.4900.

5. RESULTS

Model	Val F2	Val PR	Test F2
random_forest	1.0000	1.0000	0.9950
gradient_boosting	1.0000	1.0000	0.9799
hist_gradient_boosting	1.0000	1.0000	0.9799
adaboost	1.0000	1.0000	0.9799
voting	1.0000	1.0000	0.9799
stacking	1.0000	1.0000	0.9950
svm_rbf	0.9804	0.9820	0.9406
decision_tree	0.9804	0.9771	0.9799

6. EXPLAINABILITY AND FAIRNESS

Permutation importance ranks transformed validation features by average-precision decrease after shuffling. This is an association diagnostic, not a causal explanation. Neutral claim narratives report the score band, top signals, caveats, and next steps without alleging claimant intent. Fairness tables audit gender, age, claim type, and employment slices using recall, FPR, precision, accuracy, and predicted fraud rate; small cells are marked unstable.

7. DISCUSSION

Traditional tree and boosting methods are attractive for tabular fraud triage because they capture nonlinear interactions without a GPU and can be inspected with feature importance. However, the supplied workbook is only 4,500 rows, lacks policy and document evidence, and exhibits a highly concentrated cluster field. A future study should use license-cleared Indian claims, time-based external validation, provider history windows, and a document-aware agent system. We therefore interpret the leaderboard as a baseline under this protocol, not a universal ranking.

8. ETHICS AND LIMITATIONS

The system is decision support. Automatic denial is outside scope. A real implementation must protect claimant data, provide specific reasons, allow correction and appeal, and monitor group error rates. The workbook has no verified Indian geography or policy semantics, so localization claims are limited. Sensitive attributes are used for auditing and excluded from direct model identifiers.

9. CONCLUSION

This paper provides a reproducible traditional-ML fraud-screening baseline and a complete artifact chain. The design makes the selection rule, leakage controls, threshold, metrics, fairness audit, and limitations inspectable. Later deep and agentic approaches can use the same split and evaluator to make a fair comparison.

10. ERROR ANALYSIS AND ROBUSTNESS

A fraud screen should be reviewed through its errors rather than its headline score alone. False positives can represent legitimate high-cost or unusual claims, while false negatives can represent subtle patterns that resemble legitimate activity. The artifact contract therefore preserves confusion matrices, threshold sweeps, class support, and model probabilities. A future error-analysis run should stratify these cases by claim amount, income context, age band, specialty, claim type, submission method, and supplied cluster. These slices are descriptive and should not be interpreted as demographic causation.

Robustness is limited by the supplied sample. A random stratified split can overstate generalization when the data-generation mechanism is stable across time. The next evaluation should add a chronological holdout and an external provider or hospital split. Sensitivity should be measured after removing the cluster field, after restricting to low-cardinality features, and after introducing realistic missing documents. Those experiments would distinguish durable fraud signals from convenient synthetic shortcuts.

11. REPRODUCIBILITY AND ARTIFACT GOVERNANCE

Reproducibility is implemented as a chain rather than a sentence. The input workbook is checksummed; the configuration is hashed; the split memberships are persisted; the feature registry records transformed columns; the evaluator writes metric JSON and curve CSV files; and the report builders consume the same context. The selected model is stored with its preprocessor, threshold, target semantics, and run identifier. A new source or configuration requires a new run rather than an in-place edit of the final numbers.

This governance is important for a student project because it makes examiner questions answerable. A number in the PDF can be followed to the leaderboard, then to the model JSON, then to the source curve or threshold table. The presentation notes include the same path. A failed model is retained in the run record, so the benchmark does not quietly erase negative results.

12. INDIAN-CONTEXT DATA REQUIREMENTS

A serious Indian insurance system needs policy terms, family-floater membership, waiting periods, co-pay, sum insured, room-rent limits, hospital network status, cashless or reimbursement mode, scheme type, hospital tier, diagnosis and procedure descriptions, bills, discharge summaries, and a verified geography hierarchy. It also needs documentation and language accessibility for diverse policyholders. None of those fields are fabricated here. They are the next data-contract version for the deep-learning and agentic pillars.

Contextualization also requires care in interpretation. A higher claim amount in a metro hospital can be legitimate; an elderly claimant can have high utilization without being suspicious; a maternity or pediatric claim must not be treated as anomalous merely because it is less frequent. Fairness checks should be performed on an adequately sampled, license-cleared dataset with human review and appeal outcomes.

13. THREATS TO VALIDITY

The main threats are dataset size, uncertain provenance, possible synthetic label construction, near-unique identifiers, and a single random split. The perfect or near-perfect validation ranking of several models is a warning to investigate the data-generating process rather than a reason to claim state-of-the-art performance. The report treats this behavior as a limitation and highlights the cluster shortcut risk. Statistical tests on one split cannot repair a non-representative dataset.

14. CONCLUSION OF THE MANUSCRIPT

The contribution is a transparent baseline with a fair benchmark harness, explicit leakage controls, validation-only thresholding, locked test evaluation, calibration, explainability, fairness auditing, and submission artifacts. Its numbers are useful as an anchor for later approaches only when the same data split and evaluator are reused. The system is designed to support investigators and policyholders, not to make opaque automatic denials.

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